SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 1– Pseudocode
Weightage:	30 % of CIA	**Total Marks:**	100
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
Student Name:	P. Sudharshan Reddy	**Reg. No.:**	192372204
Section / Batch:	CSE(AI)	**Date:**	24-10-2026

Code of Conduct

I P.Sudharshan Reddy(192372204) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Sudharshan Reddy

Instructions

- Duration: 1 hour
- Answer all Questions
- Write Pseudocode in paper and upload program in GITHUB

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____

Questions

1. Design a health and self-care mobile app that helps users maintain mental wellbeing with meditations and exercises. This case study talks you through the design process of creating a user- friendly mobile app. Design UI with the following requirements.
   - User Login
   - Register
   - Health Routine Blogs
   - Self-Care Blogs

2. This is an intelligent job assistant that helps users find their ideal job and get to all the way to signing a contract faster and more easily. Design UI with the following requirements.
   - User Login
   - Register Subscription
   - Personal Detail
   - Resume Upload
   - Choosing Job

3. This is a mobile app that can help people get parking slots easily even when they travel beyond their normal routes. Design UI with the following requirements.
   - UserLogin
   - Register Count of Empty Parking
   - Free Parking Slots Display
   - Booking Parking Slots

4. Design a mobile app for Online Body Fitness tips and videos for a good daily routine. Contains the following requirements.

   User Login
   - Register
   - Fitness Blog
   - Diet Planner
   - Payment and Subscription

ITA0307 - Mobile Computing
C03 - AT1
PSEUDO CODE WRITING

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CSE - A1

① Mental Wellbeing app.

```
Screen login
  Input email, password
  If fields empty then Show "fill all fields"
  else if Credentials valid THEN Save Session
  ELSE Show "Invalid Credentials"
SCREEN Register
  INPUT name, email, password, Confirm password
  IF invalid fields THEN show error
  ELSE Save user → GOTO Login
SCREEN Home
  Display HealthRoutineBlogs, SelfCareBlogs
SCREEN HealthRoutine Blogs
  Display health routine Blog Cards
  ON tap → GOTO BlogDetail
  SCREEN SelfCare Blogs
  DISPLAY Self care blog cards
  ON tap → GOTO BlogDetail.
END
```

② Intelligent Job Assistant

SCREEN LOGIN

INPUT email, password

IF valid THEN GOTO Dashboard

ELSE show error

SCREEN RegisterSubscriptPay

INPUT name, email, password

SELECT plan

IF Premium/Pro THEN process payment

IF payment successfull THEN save User

GOTO PersonalDetail

SCREEN PersonalDetail

INPUT personal and education details

IF fields empty THEN show error

ELSE save profile → GOTO ResumeUpload.

SCREEN ChoosingJob

GET all jobs. CALCULATE skill match score

SORT jobs by score Desc DISPLAY Jobs

ON Apply → save application

SHOW "Application SENT"

END

③ Parking Slot Finder

```
SCREEN UserLogin
    INPUT email, password
    IF valid THEN GOTO Dashboard
    ELSE show error

SCREEN RegisterEmptyParking
    INPUT lotName, address, totalSlots, emptySlots, price
    IF emptySlots > totalSlots THEN show error
    ELSE save parking lot -> show Confirmation

SCREEN FreeParkingDisplay
    GET user location
    GET nearby slots with emptySlots > 0
    SORT lots By distance, slots, price
    ON lot tap -> GOTO BookingParkingSlot

SCREEN BookingParkingSlot
    INPUT VehicleNumber, StartTime, duration
    IF no slots THEN show "Parking full"
    ELSE Calculate Cost
        ON Confirm:
            UPDATE available slots
            SAVE booking and process payment
ON booking  DISPLAY Confirmation and QR Code
 END
```

① ONLINE Bodyfitnelf -App

SCREEN USERLOGIN

INPUT email. password
IF Valid THEN GOTO Home
ELSE Show error

SCREEN Register

INPUT user details and fitness goals
IF email exists THEN Show error
ELSE Save User → GOTO Login

SCREEN Diet planner

INPUT goal, dietType, allergies
CALCULATE Daily Calories
GENERATE meal plan DISPLAY meals
ON Save - → Save plan.

SCREEN paymentAnd Subscription.

SELECT subscription plan
INPUT payment details
IF payment Succelfull THEN
ACTIVATE premium plan
SET expiryDate
ELSE Show "payment Failed"

DAILY CHECK
IF Today > expiryDate THEN SET plan = Free
END